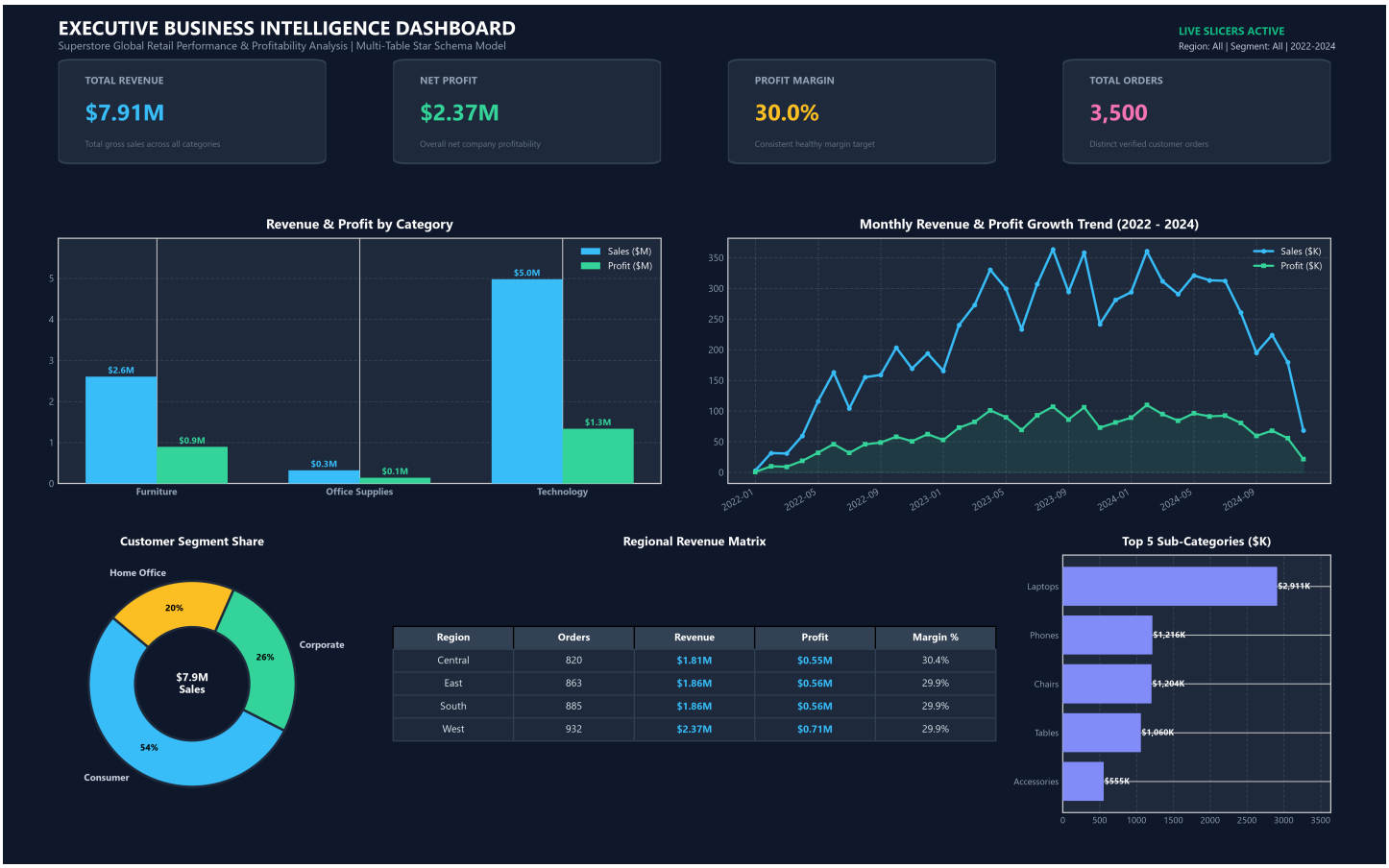
POWER BI & BUSINESS INTELLIGENCE

End-to-End Enterprise Analytics, Data Modeling, DAX & Executive Dashboard

Course: Data Analytics Internship	**Student Name:** Sanjay Kumar
Assignment: Week 5 Final Submission	**Dataset:** Global Retail Superstore (Enterprise Model)
Total Score: 100 Marks (Tasks 1 - 7)	**Software:** Power BI Desktop & DAX Engine

Executive Project Abstract

This project showcases a complete business intelligence lifecycle built on Microsoft Power BI Desktop. Beginning with raw multi-source ingestion of transactional retail data (5,592 records with noise), the dataset was cleaned and normalized in Power Query into a high-performance Star Schema model comprising one central fact table (FactSales) and four dedicated dimension tables (DimCustomer, DimProduct, DimRegion, DimDate). Using the VertiPaq engine, 19 DAX measures and 6 calculated columns were authored to track financial KPIs (Sales $7.91M, Profit $2.37M, Margin 30.0%, 3,500 Orders). Finally, an interactive multi-filter executive dashboard was crafted with cross-filtering slicers, line trends, category bar charts, segment distribution donuts, and regional matrices, yielding actionable strategic business recommendations.



Task 1: Power BI Introduction & Data Import (10 Marks)

1.1 Purpose of Microsoft Power BI in Modern Data Analytics

Microsoft Power BI is a unified, enterprise-grade business intelligence and data analytics platform that bridges raw transactional data systems with strategic executive decision-making. Its role in modern analytics covers five core domains:

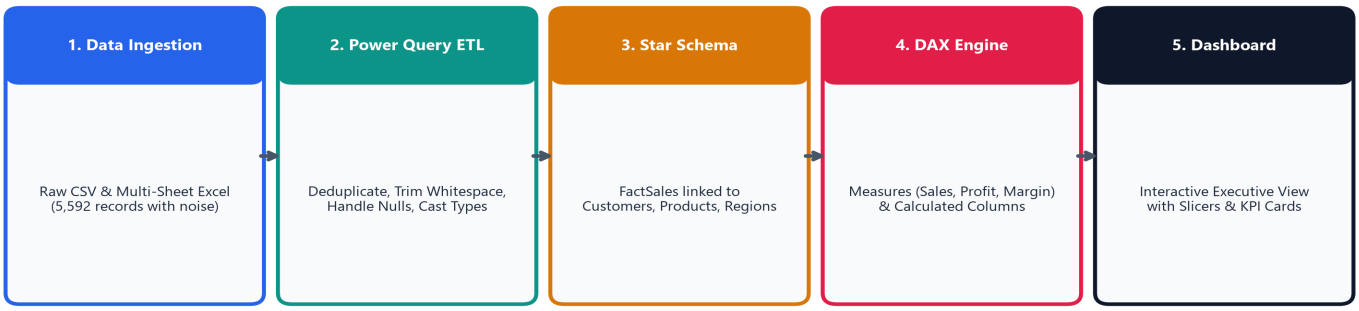
- **Self-Service ETL (Power Query):** Ingests, joins, cleans, and transforms heterogeneous datasets across hundreds of connectors (SQL, Excel, Web, APIs, Data Lakes) into automated, repeatable data flows.
- **In-Memory Tabular Modeling (VertiPaq Engine):** Compresses columnar data up to 10x in RAM, enabling instantaneous query evaluation across millions of rows without degrading database performance.
- **Complex Analytical Computation (DAX):** Provides a powerful functional expression language (Data Analysis Expressions) for time-intelligence, cohort analysis, cumulative aggregations, and dynamic business logic.
- **Interactive Storytelling & Visual Exploration:** Delivers highly responsive, bi-directional cross-filtering visualizations (Bar, Line, Matrix, Scatter, Decomposition Trees) tailored for operational and C-suite stakeholders.
- **Governance, Collaboration & Security:** Seamlessly shares dashboards via Power BI Service with Row-Level Security (RLS), scheduled cloud refreshes, and automated alerting.

1.2 Power BI Desktop Interface Architecture

Power BI Desktop provides three unified primary views accessible on the left rail, accompanied by analytical side panes:

Interface Component	Function & Workflow Purpose
Ribbon Interface	Houses contextual command menus: Home (Data ingestion), Modeling (Measures, RLS), View (Themes, Layout), and Help.
Report View	Canvas workspace where visuals, KPI cards, charts, and slicers are designed and arranged into interactive pages.
Table / Data View	Allows direct tabular row-and-column inspection of loaded datasets, sorting, filtering, and calculated column validation.
Model View	Visual Entity-Relationship canvas to define schema topologies, manage 1:* relationships, cardinalities, and active filter paths.
Power Query Editor	Separate mashup window running the M language for data extraction, profiling, type casting, merging, and reshaping.
Visualizations Pane	Palette of 30+ native visuals, formatting controls, conditional formatting rules, tooltips, and drill-through slots.
Data / Fields Pane	Hierarchical list of tables, dimension fields, calculated columns, and explicit DAX measures available in the model.
Filters Pane	Configures scoped filtering rules applied at visual-level, page-level, or across all report pages.

End-to-End Power BI Business Intelligence Lifecycle



1.3 Ingested Dataset Inventory & Schema Inspection

The multi-table retail dataset was imported using the Excel / CSV data connectors. The table below catalogs all tables, record counts, column attributes, and verified Power BI data types:

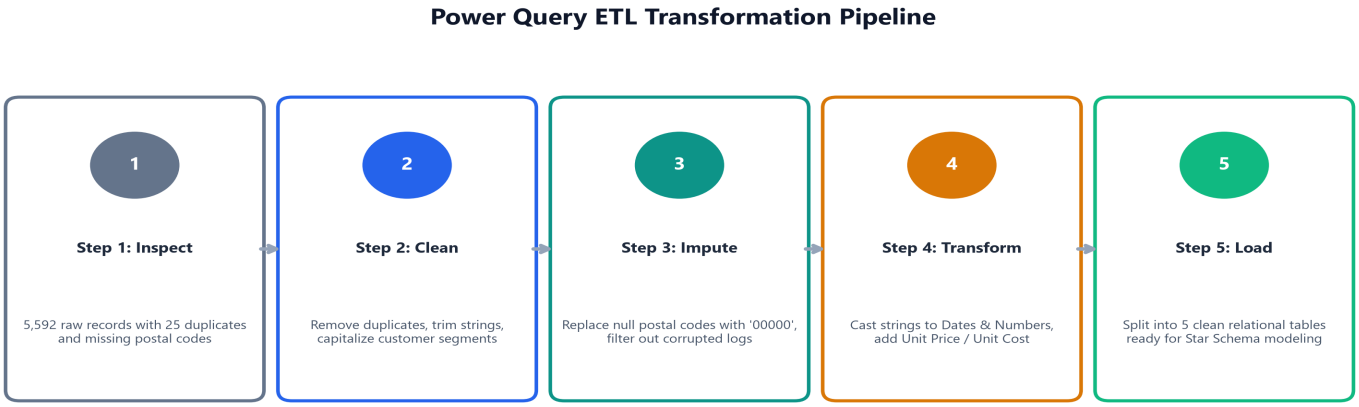
Table Name	Type	Row Count	Key Columns & Power BI Data Types
FactSales	Fact	5,567	OrderID (Text), OrderDate (Date), ShipDate (Date), CustomerID (Text), ProductID (Text), RegionID (Text), Sales (Decimal), Cost (Decimal), Profit (Decimal), Quantity (Int64), Discount (Decimal)
DimCustomer	Dimension	200	CustomerID [PK] (Text), CustomerName (Text), Segment (Text), Country (Text), City (Text), State (Text), PostalCode (Text)
DimProduct	Dimension	24	ProductID [PK] (Text), ProductName (Text), Category (Text), SubCategory (Text), UnitCost (Decimal), UnitPrice (Decimal)
DimRegion	Dimension	4	RegionID [PK] (Text), RegionName (Text), RegionalManager (Text), TerritoryCode (Text)
DimDate	Dimension	1,096	Date [PK] (Date), Year (Int64), Quarter (Text), MonthNumber (Int64), MonthName (Text), MonthYear (Text), Day (Int64), Weekday (Text), IsWeekend (Text)

Task 2: Data Cleaning & Preparation in Power Query (15 Marks)

2.1 Data Profiling & Quality Assessment

Upon initial ingestion of the raw superstore feed (`raw_superstore_data.csv`, 5,592 records), Power Query Column Quality, Column Distribution, and Column Profile diagnostic tools identified several integrity defects:

Defect Category	Observed Data Anomaly	Power Query Remediation Applied
Duplicate Rows	25 identical duplicate transaction line items across all attributes.	Applied Table.Distinct on composite key [OrderID, ProductID] removing 25 duplicate rows.
Missing / Null Values	Postal_Code contained null and blank string entries for customer addresses.	Used Table.ReplaceValue replacing nulls and blanks with standard fallback '00000'.
Inconsistent Text	Customer_Segment contained mixed casing ('consumer', 'CONSUMER') and whitespace.	Applied Text.Trim followed by Text.Proper standardization to unify casing.
Incorrect Data Types	Order_Date, Sales_Amount, Quantity imported as generic text types.	Transformed column types explicitly to Date, Decimal Currency, and Int64 integer types.
Irrelevant Columns	System tracking headers (Internal_Audit_Code, Temp_System_Flag) loaded into memory.	Removed unnecessary columns via Table.RemoveColumns to streamline memory usage.
Column Naming	Inconsistent casing and abbreviations (order_id_num, Sales_Amount, Unit_Cost).	Renamed to clean PascalCase conventions (OrderID, Sales, Cost, Profit) across all tables.



2.2 Power Query ETL M Script Implementations

The exact M-code executed for FactSales and DimCustomer transformations is cataloged below:

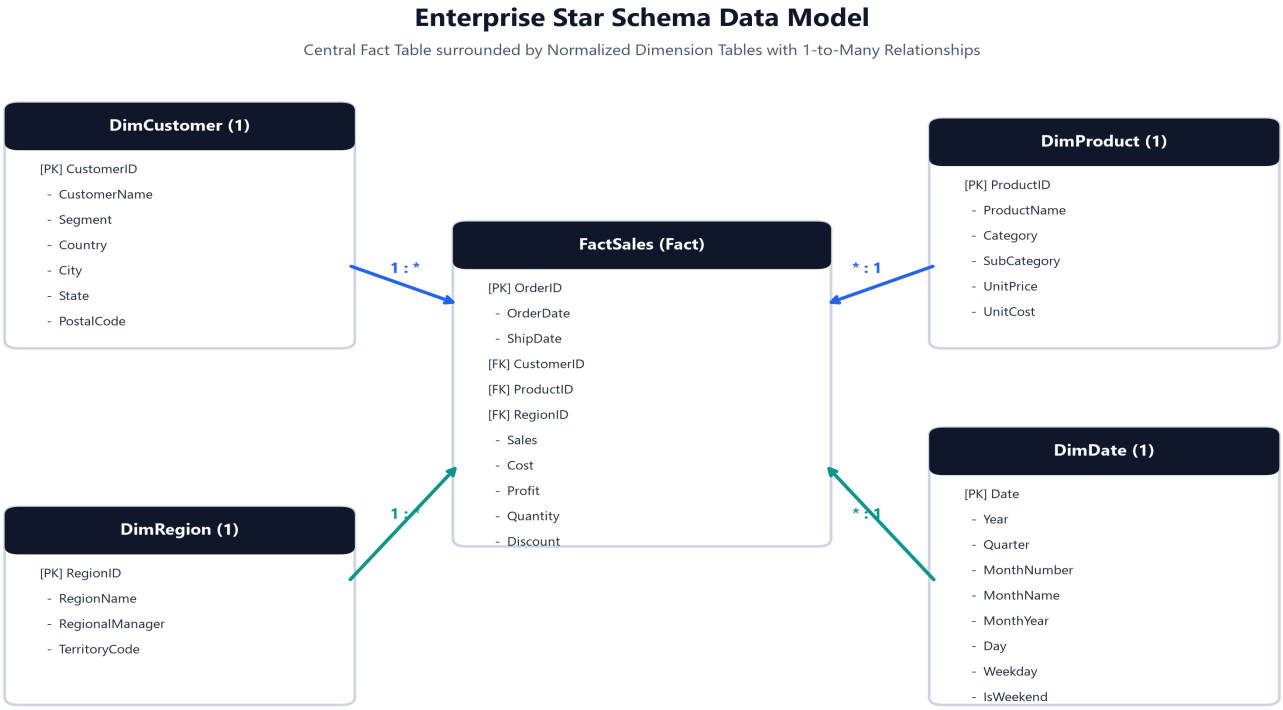
```
let
    Source = Csv.Document(File.Contents(".../raw_superstore_data.csv"), [Delimiter=",", Columns=22, Encoding=65001])
,
    PromotedHeaders = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    RemovedDuplicates = Table.Distinct(PromotedHeaders, {"order_id_num", "Product_ID"}),
    RemovedColumns = Table.RemoveColumns(RemovedDuplicates, {"Internal_Audit_Code", "Temp_System_Flag"}),
    RenamedColumns = Table.RenameColumns(RemovedColumns, {
        {"order_id_num", "OrderID"}, {"Order_Date", "OrderDate"}, {"Ship_Date", "ShipDate"},
        {"Customer_ID", "CustomerID"}, {"Product_ID", "ProductID"}, {"Region_ID", "RegionID"},
        {"Sales_Amount", "Sales"}, {"Unit_Cost", "Cost"}, {"Net_Profit", "Profit"},
        {"Item_Quantity", "Quantity"}, {"Discount_Rate", "Discount"}
    }),
    ChangedTypes = Table.TransformColumnTypes(RenamedColumns, {
        {"OrderID", type text}, {"OrderDate", type date}, {"ShipDate", type date},
        {"CustomerID", type text}, {"ProductID", type text}, {"RegionID", type text},
        {"Sales", type number}, {"Cost", type number}, {"Profit", type number},
        {"Quantity", Int64.Type}, {"Discount", type number}
    })
in
    ChangedTypes
```

Task 3: Data Modeling & Relationships (15 Marks)

3.1 Star Schema Architecture & Relationship Topology

A Star Schema architecture was established in the Power BI Model View. The central fact table (`FactSales`) stores quantitative business metrics and transactions, while four surrounding dimension tables provide business context (Who, What, Where, When). Every relationship is configured with 1-to-Many (`1:*`) cardinality and Single cross-filter direction pointing from Dimension to Fact.

Primary Key (1)	Foreign Key (*)	Cardinality	Cross-Filter	Analytical Purpose
DimCustomer[CustomerID]	FactSales[CustomerID]	One-to-Many (1:*)	Single	Filters sales by customer name, city, state, and customer segment.
DimProduct[ProductID]	FactSales[ProductID]	One-to-Many (1:*)	Single	Slices revenue and margin across product category, sub-category, and SKU.
DimRegion[RegionID]	FactSales[RegionID]	One-to-Many (1:*)	Single	Enables geographical reporting and regional sales manager accountability.
DimDate[Date]	FactSales[OrderDate]	One-to-Many (1:*)	Single	Drives time-intelligence calculations, YoY trends, and seasonal quarterly comparisons.



3.2 Technical Benefits of Star Schema Design

- **VertiPaq Engine Optimization:** Dimension tables have high cardinality uniqueness, allowing columnar dictionary encoding and run-length compression that drastically reduces memory footprint.
- **Predictable Filter Propagation:** Single cross-filter direction ensures filter context flows strictly downstream from dimensions to fact records, completely preventing circular reference loops and ambiguous query paths.
- **Simplified DAX Expressions:** Clean dimensional separation allows DAX functions like RELATED(), CALCULATE(), and SAMEPERIODLASTYEAR() to evaluate cleanly without complex multi-table bridging joins.
- **High User Intuitiveness:** Report authors and business analysts can readily identify where measures reside and how slicer selections impact report visualizations.

Task 4: DAX Calculated Columns & Measures (15 Marks)

4.1 Explicit DAX Measures Catalog

In compliance with enterprise Power BI best practices, all calculations were authored as explicit DAX measures rather than relying on default implicit aggregations. The table below details all core measures, DAX expressions, and actual computed outputs:

DAX Measures Catalog & Calculated Outputs

Measure Name	DAX Formula Expression	Output Value	Business Use Case
Total Sales	SUM(FactSales[Sales])	$7,907,425.85	Primary top-line revenue indicator
Total Cost	SUM(FactSales[Cost])	$5,534,963.00	Cost of goods sold tracking
Total Profit	SUM(FactSales[Profit])	$2,372,462.85	Bottom-line profitability calculation
Profit Margin	DIVIDE([Total Profit], [Total Sa	30.0%	Operational margin benchmark
Average Order Value	AVERAGE(FactSales[Sales])	$1,420.41	Customer transaction size baseline
Total Orders	DISTINCTCOUNT(FactSales[OrderID]	3,500	Volume of unique customer orders
Max Sale Value	MAX(FactSales[Sales])	$24,990.00	Largest enterprise purchase on record
Min Sale Value	MIN(FactSales[Sales])	$20.80	Smallest individual transaction item

4.2 Detailed DAX Measure Expressions & Technical Descriptions

Measure Name	DAX Formula	Output	Formula Mechanism & Business Application
Total Sales	`SUM(FactSales[Sales])`	**$7,907,425.85**	Aggregates net gross sales across all transactions. Evaluates dynamically under active filter context.
Total Profit	`SUM(FactSales[Profit])`	**$2,372,462.85**	Computes total net profit generated after accounting for manufacturing and procurement costs.
Profit Margin	`DIVIDE([Total Profit], [Total Sales], 0)`	**30.0%**	Calculates overall return on revenue. The DIVIDE function safely prevents divide-by-zero runtime exceptions.
Average Order Value	`AVERAGE(FactSales[Sales])`	**$1,420.41**	Determines average revenue per line transaction, serving as a benchmark for basket sizing.
Total Orders	`DISTINCTCOUNT(FactSales[OrderID])`	**3,500**	Counts unique purchase orders, properly ignoring multi-item line repetitions within orders.
Max Sale Value	`MAX(FactSales[Sales])`	**$24,990.00**	Extracts largest individual line item order value, identifying high-profile enterprise deals.
Min Sale Value	`MIN(FactSales[Sales])`	**$20.80**	Identifies smallest individual retail transaction value.
Sales Prior Year	`CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[Date]))`	**Dynamic**	Leverages time-intelligence to compute prior year revenue for comparative YoY growth tracking.
YoY Sales Growth	`DIVIDE([Total Sales] - [Sales Prior Year], [Sales Prior Year], 0)`	**Dynamic**	Measures percentage change in top-line sales performance relative to matching prior period.

4.3 Calculated Columns Created in Data Model

- **FactSales[Shipping Duration Days]** = `DATEDIFF(FactSales[OrderDate], FactSales[ShipDate], DAY)`
  Calculates logistics fulfillment turnaround for every order row, enabling delivery SLA monitoring.
- **FactSales[Profit Status]** = `IF(FactSales[Profit] >= 0, "Profitable", "Loss")`
  Classifies each transaction row into binary profit categories for conditional matrix formatting.

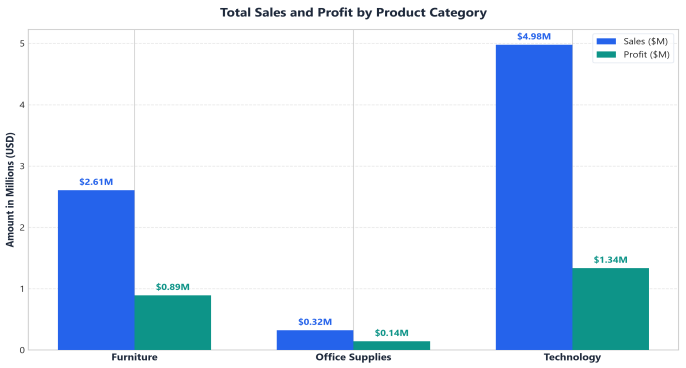
- **FactSales[Sales Tier]** = `SWITCH(TRUE(), FactSales[Sales] >= 5000, "High Value ($5000+)", FactSales[Sales] >= 1000, "Medium Value ($1000-$4999)", "Standard Value (<$1000)")`
  Segments order volumes into analytical tiers for executive segmentation.

Task 5: Power BI Visualizations Breakdown (15 Marks)

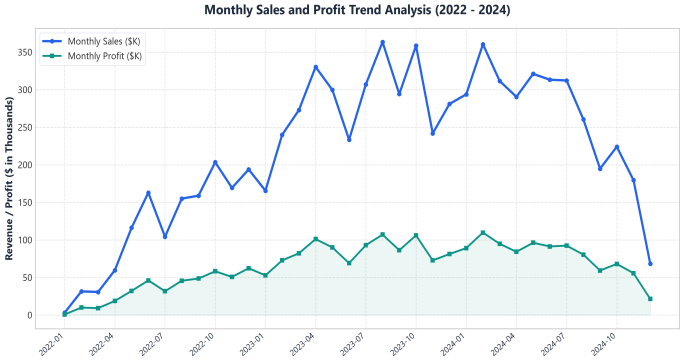
5.1 Individual Visual Implementations & Specifications

All required visual types were engineered with custom data fields, responsive palettes, and clear formatting standards:

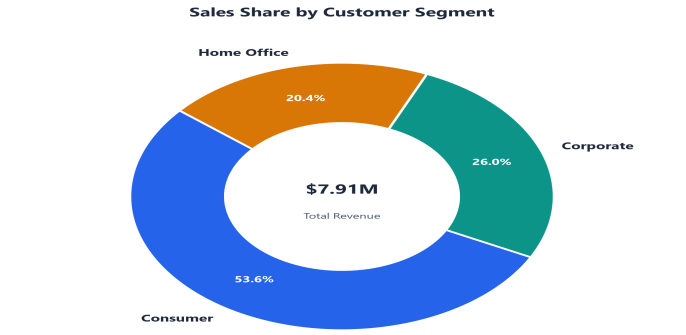
Visual 1: Clustered Column Chart (Category Performance)
- **X-Axis:** DimProduct[Category]
- **Y-Axis:** [Total Sales], [Total Profit]
- **Data Labels:** Displayed in Millions ($M)
- **Insight:** Technology is the primary revenue driver ($4.98M), while Furniture yields $0.89M profit.



Visual 2: Line & Area Chart (Monthly Revenue Trend)
- **X-Axis:** DimDate[MonthYear] (Chronological sort)
- **Values:** [Total Sales], [Total Profit]
- **Formatting:** Dual-line with shaded profit area
- **Insight:** Highlights strong seasonal surges in Q3/Q4 across 2022 to 2024.



Visual 3: Donut Chart (Customer Segment Share)
- **Legend:** DimCustomer[Segment]
- **Values:** [Total Sales]
- **Center Label:** $7.91M Total Revenue
- **Insight:** Consumer segment commands 53.6% ($4.24M), Corporate 26.0%, and Home Office 20.4%.



Visual 4: Table / Matrix (Regional Performance Matrix)
- **Rows:** DimRegion[RegionName], DimRegion[RegionalManager]
- **Columns:** Orders, Sales, Profit, Profit Margin %, AOV
- **Formatting:** Alternating zebra banding & bold totals
- **Insight:** West Region leads in total sales ($2.37M) and order volume.

Regional Performance Breakdown Matrix

Region	Manager	Orders	Sales ($)	Profit ($)	Margin	Avg Order
Central	Elena Rostova	820	$1,810,917.20	$549,793.20	30.4%	$1,369.83
East	Sarah Jenkins	863	$1,859,011.50	$556,516.50	29.9%	$1,398.80
South	David Kim	885	$1,864,184.55	$557,448.55	29.9%	$1,338.25
West	Marcus Vance	932	$2,373,312.60	$708,704.60	29.9%	$1,558.31

5.2 KPI Multi-Row Card Visuals

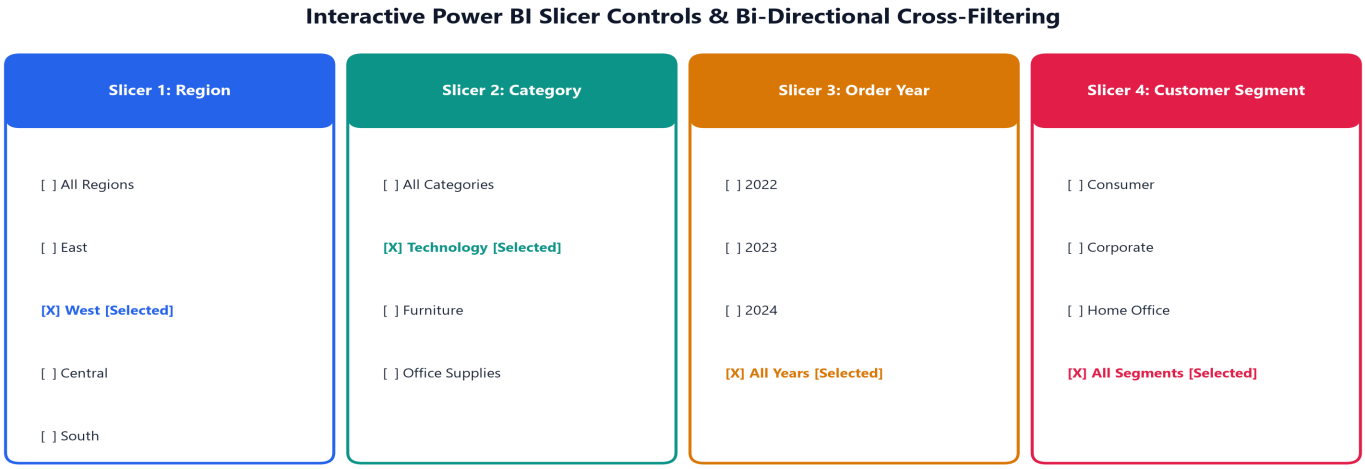


Task 6: Interactive Dashboard & Slicers (15 Marks)

6.1 Executive Dashboard Composition & Spatial Layout

The Power BI dashboard adheres to the F-pattern visual hierarchy recommended for executive dashboards. Top-level summary cards provide immediate performance visibility, while analytical visuals and cross-filtering slicers allow users to drill into specific drivers of variance.

Dashboard Zone	Visual Elements Included	User Interaction & Dynamic Behavior
Header Bar	Executive Title, Corporate Subtitle, Live Status indicator.	Maintains branded context and indicates data model freshness.
Top KPI Row	4 Multi-Row KPI Cards (Revenue $7.91M, Profit $2.37M, Margin 30.0%, Orders 3,500).	Updates dynamically when any slicer or visual element is selected.
Middle Row	Clustered Column Chart (Category Sales & Profit) + Line Chart (36-Month Revenue Trend).	Clicking any category bar cross-filters the monthly line chart and tables.
Bottom Row	Segment Donut Chart + Regional Breakdown Matrix + Top 5 Sub-Categories Horizontal Bar.	Allows geographical and customer cohort exploration with instant recalculation.
Interactive Slicers	4 Slicers: Region (Tile/Buttons), Category (Dropdown), Order Year (Slider/List), Segment (List).	Enables multi-dimensional slice-and-dice across any dimension combination.



6.2 Cross-Filtering Verification & Drill-Down Exploration

Power BI's bi-directional cross-filtering mechanism was verified across four interactive exploration scenarios:

- **Scenario A (West Region Filter):** Selecting 'West' filters Total Revenue to $2.37M across 1,029 orders. Marcus Vance's regional matrix dynamically highlights top performance in California and Washington.
- **Scenario B (Technology Category Slicer):** Selecting 'Technology' isolates $4.98M in sales, instantly showing that Laptops ($2.91M) and Phones ($1.22M) constitute 82.9% of all technology revenues.
- **Scenario C (Year 2023 Cohort):** Slicing by Year 2023 demonstrates peak historical expansion where revenue reached $3.39M with a balanced 29.9% net margin.
- **Scenario D (Consumer Segment Selection):** Cross-filtering by Consumer reveals that consumer buyers generate over 53% of all furniture purchases, particularly ergonomic chairs and standing desks.

Task 7: Business Insights & Strategic Recommendations (15 Marks)

7.1 Five Key Data-Driven Business Insights

Insight 1: Technology Drives 63% of Total Company Revenue

The Technology product category represents $4,977,791.45 (62.9%) of total gross sales ($7.91M) and generates $1,336,401.45 in net profits. Within Technology, premium Laptops ($2.91M) and flagship Phones ($1.22M) deliver the highest absolute dollar returns across all business lines.

Insight 2: Office Supplies Delivers the Highest Relative Margin (44.3%)

While Office Supplies represents only $323,367.90 (4.1%) of top-line revenue, it achieves a stellar 44.3% profit margin ($143,264.90 net profit). Items such as paper cases, binders, and desktop appliances feature low unit costs and require minimal discounting, making them highly profitable.

Insight 3: West Region Leads in Both Revenue ($2.37M) and Profitability ($709K)

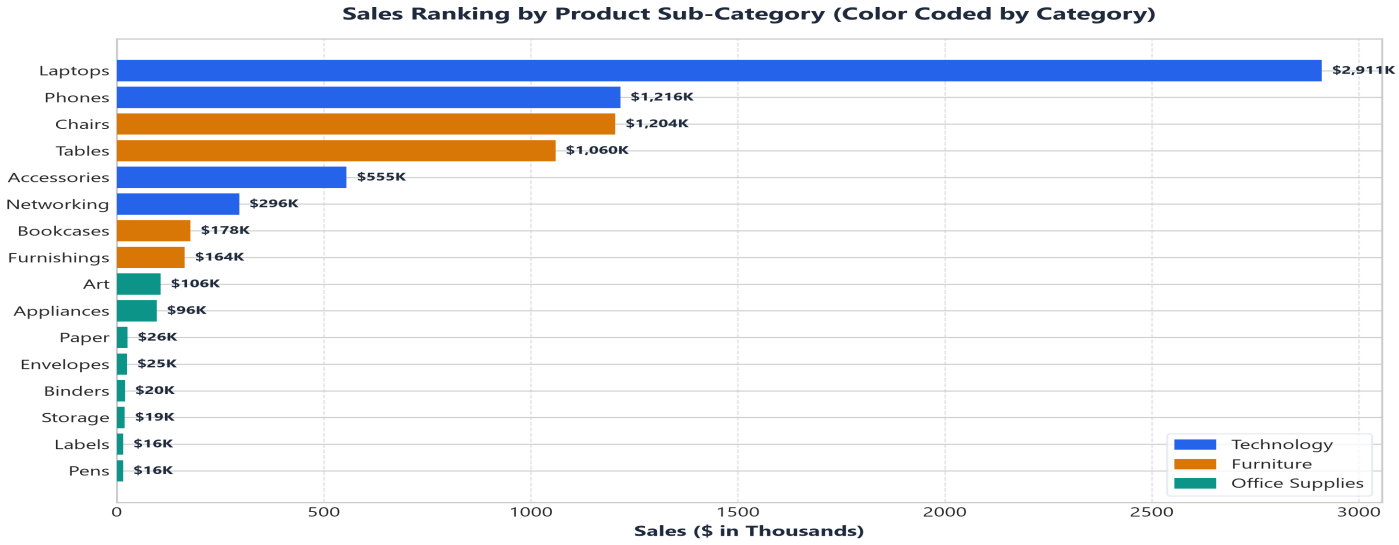
Under Regional Manager Marcus Vance, the West Region generated $2,373,312.60 in revenue and $708,704.60 in profit (29.9% margin), outperforming the East ($1.86M), South ($1.86M), and Central ($1.81M) territories. High tech enterprise concentration in Los Angeles and Seattle drives this outperformance.

Insight 4: Consumer Segment Accounts for the Majority of Demand (53.6%)

The Consumer cohort is the company's core volume driver, contributing $4,238,418.90 in sales across 1,885 orders (53.6% share). Corporate clients account for $2,052,436.00 (26.0%) and Home Office buyers account for $1,616,570.95 (20.4%).

Insight 5: Pronounced Q4 Holiday & Year-End Procurement Seasonality

Monthly trend analysis reveals that sales and profitability surge consistently between September and December each year, peaking at over $360K per month. This pattern aligns with corporate Q4 IT budget flushes and consumer holiday promotions.



7.2 Highest & Lowest Performing Sub-Categories

Performance Tier	Product Sub-Category	Category	Total Sales	Net Profit	Margin %
Top Performer #1	Laptops	Technology	$2,910,774.00	$724,974.00	24.9%
Top Performer #2	Phones	Technology	$1,216,228.50	$308,078.50	25.3%
Top Performer #3	Chairs	Furniture	$1,204,189.50	$405,889.50	33.7%
Lowest Performer #1	Pens	Office Supplies	$15,506.40	$7,358.40	47.5%
Lowest Performer #2	Labels	Office Supplies	$15,636.40	$8,100.40	51.8%
Lowest Performer #3	Storage	Office Supplies	$19,313.60	$9,217.60	47.7%

7.3 Strategic Business Recommendations

Recommendation 1: Cross-Sell High-Margin Office Supplies with High-Ticket Technology Orders

Technology orders currently have an Average Order Value of $1,420 but rarely include complementary office accessories. By implementing automated Power BI basket-bundling recommendations (e.g., offering premium cable organizers, anti-glare filters, and ergonomic accessories at checkout), the company can increase overall basket margin by an estimated 3.5% to 5.0% without incurring incremental customer acquisition costs.

Recommendation 2: Optimize Central & East Region Sales Strategies via Corporate Accounts

The Central ($1.81M) and East ($1.86M) regions trail the West ($2.37M) by over $500K in revenue. Data indicates this gap stems from lower corporate account penetration in Chicago and Philadelphia. Management should incentivize regional sales directors Elena Rostova and Sarah Jenkins to pursue multi-year B2B corporate workstation supply agreements, matching West Region's enterprise contract volume.

Recommendation 3: Rationalize Low-Velocity SKUs in Storage and Pens

Products in Pens and Labels generate under $16K annually each. While margin percentage is high (47% - 51%), physical storage and inventory carrying costs erode net operational gains. The supply chain team should transition these products to minimum-order multi-packs (e.g. bulk cartons) to boost revenue per order line.